Assignment No. 9: Logic Synthesis using Synopsys Design Compiler

Q1. Design and synthesize a 4-bit Universal Shift Register using Verilog HDL.
a) Implement all functional modes of operation.
b) Apply appropriate design constraints to achieve a synthesizable implementation.
c) Verify the functional correctness of the synthesized design.
d) Interpret the generated synthesis results with respect to area, timing, and power.

- ## Testbench:

```
`timescale 1ns/1ps
module tb_universal_shift_register;
    reg clk, rst;
    reg [1:0] mode;
    reg [3:0] parallel_in;
    reg serial_in_left, serial_in_right;
    wire [3:0] q;
    universal_shift_register uut (
        .clk(clk),
        .rst(rst),
        .mode(mode),
        .parallel_in(parallel_in),
        .serial_in_left(serial_in_left),
        .serial_in_right(serial_in_right),
        .q(q)
    );
    always #5 clk = ~clk;
    initial begin
        $dumpfile("universal_shift_register.vcd");
        $dumpvars(0, tb_universal_shift_register);

        clk = 0;
        rst = 1;
        mode = 2'b00;
        parallel_in = 4'b0000;
        serial_in_left = 0;
        serial_in_right = 0;
        #12 rst = 0;
        // Parallel load: Q = 1010
        @(negedge clk);
        mode = 2'b11;
        parallel_in = 4'b1010;
```

```
        // Hold
        @(negedge clk);
        mode = 2'b00;

        // Shift right
        @(negedge clk);
        mode = 2'b01;
        serial_in_left = 1;

        // Shift left
        @(negedge clk);
        mode = 2'b10;
        serial_in_right = 1;

        // Load a different value
        @(negedge clk);
        mode = 2'b11;
        parallel_in = 4'b1100;

        // Hold again
        @(negedge clk);
        mode = 2'b00;

        repeat (2) @(negedge clk);

        $finish;
    end
    initial begin
        $monitor("Time=%0t rst=%b mode=%b parallel_in=%b q=%b",
                 $time, rst, mode, parallel_in, q);
    end
endmodule
```

- ## Commands:

vcs -o Output/universal_shift_register.out ../RTL/Design/universal_shift_register.v ../RTL/TB/tb_universal_shift_register.v

cd Output
./universal_shift_register.out

source synth.tcl

gui_show

- **sdc file constraint: create_clock -name clk -period 10 [get_ports clk]**

- **Synth File:**

~/Desktop/Shashaank_Malviya/Scripts/synth.tcl - Mousepad

File Edit Search View Document Help

```
source /home/oitp_b2-13/Desktop/Shashaank_Malviya/Scripts/dc_setup.tcl

set TOP universal_shift_register

current_design $TOP

read_verilog /home/oitp_b2-13/Desktop/Shashaank_Malviya/RTL/Design/${TOP}.v


link

source /home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Constraints/${TOP}.sdc

check_design

compile_ultra

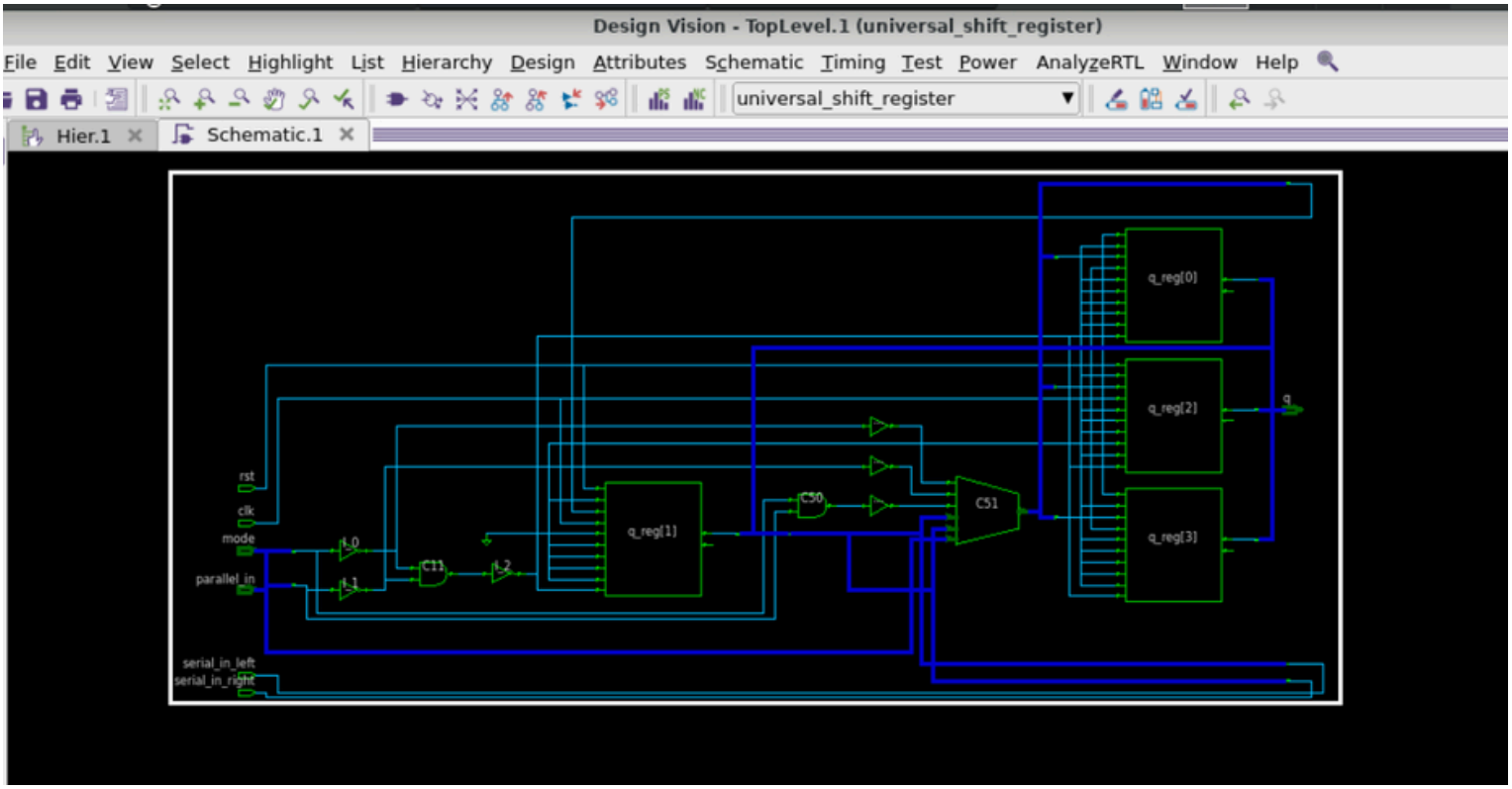
report_qor    > /home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Reports/${TOP}_qor.rpt
report_area   >/home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Reports/${TOP}_area.rpt
report_timing > /home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Reports/${TOP}_timing.rpt
report_power  > /home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Reports/${TOP}_power.rpt

write -format verilog \
-hierarchy \
-output /home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Netlist/${TOP}_synth_netlist.v

write_sdc output_sdc_for_PD.sdc

write_file \
-format ddc \
-hierarchy \
-output /home/oitp_b2-13/Desktop/Shashaank_Malviya/Synthesis/Backup/${TOP}.ddc
```

- **Schematic:**



- **Netlist:**

```
module universal_shift_register ( clk, rst, mode, parallel_in, serial_in_left,
        serial_in_right, q );
  input [1:0] mode;
  input [3:0] parallel_in;
  output [3:0] q;
  input clk, rst, serial_in_left, serial_in_right;
  wire   N0, N1, N2, N3, N4, N5, N6, N7, N8, N9, N10, N11;

  GTECH_AND2 C11 ( .A(N4), .B(N5), .Z(N3) );
  \**SEQGEN**  \q_reg[3]  ( .clear(rst), .preset(1'b0), .next_state(N10),
        .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(q[3]),
        .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0),
        .synch_enable(N11) );
  \**SEQGEN**  \q_reg[2]  ( .clear(rst), .preset(1'b0), .next_state(N9),
        .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(q[2]),
        .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0),
        .synch_enable(N11) );
  \**SEQGEN**  \q_reg[1]  ( .clear(rst), .preset(1'b0), .next_state(N8),
        .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(q[1]),
        .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0),
        .synch_enable(N11) );
  \**SEQGEN**  \q_reg[0]  ( .clear(rst), .preset(1'b0), .next_state(N7),
        .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(q[0]),
        .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0),
        .synch_enable(N11) );
  GTECH_NOT I_0 ( .A(mode[1]), .Z(N4) );
  GTECH_NOT I_1 ( .A(mode[0]), .Z(N5) );
  GTECH_AND2 C50 ( .A(mode[1]), .B(mode[0]), .Z(N6) );
  SELECT_OP C51 ( .DATA1({serial_in_left, q[3:1]}), .DATA2({q[2:0],
        serial_in_right}), .DATA3(parallel_in), .CONTROL1(N0), .CONTROL2(N1),
        .CONTROL3(N2), .Z({N10, N9, N8, N7}) );
  GTECH_BUF B_0 ( .A(N4), .Z(N0) );
  GTECH_BUF B_1 ( .A(N5), .Z(N1) );
```

- **Power Report:**

```
    Time Units = Unitless
    Dynamic Power Units = Unitless    (derived from V,C,T units)
    Leakage Power Units = Unitless


Attributes
----------
i - Including register clock pin internal power

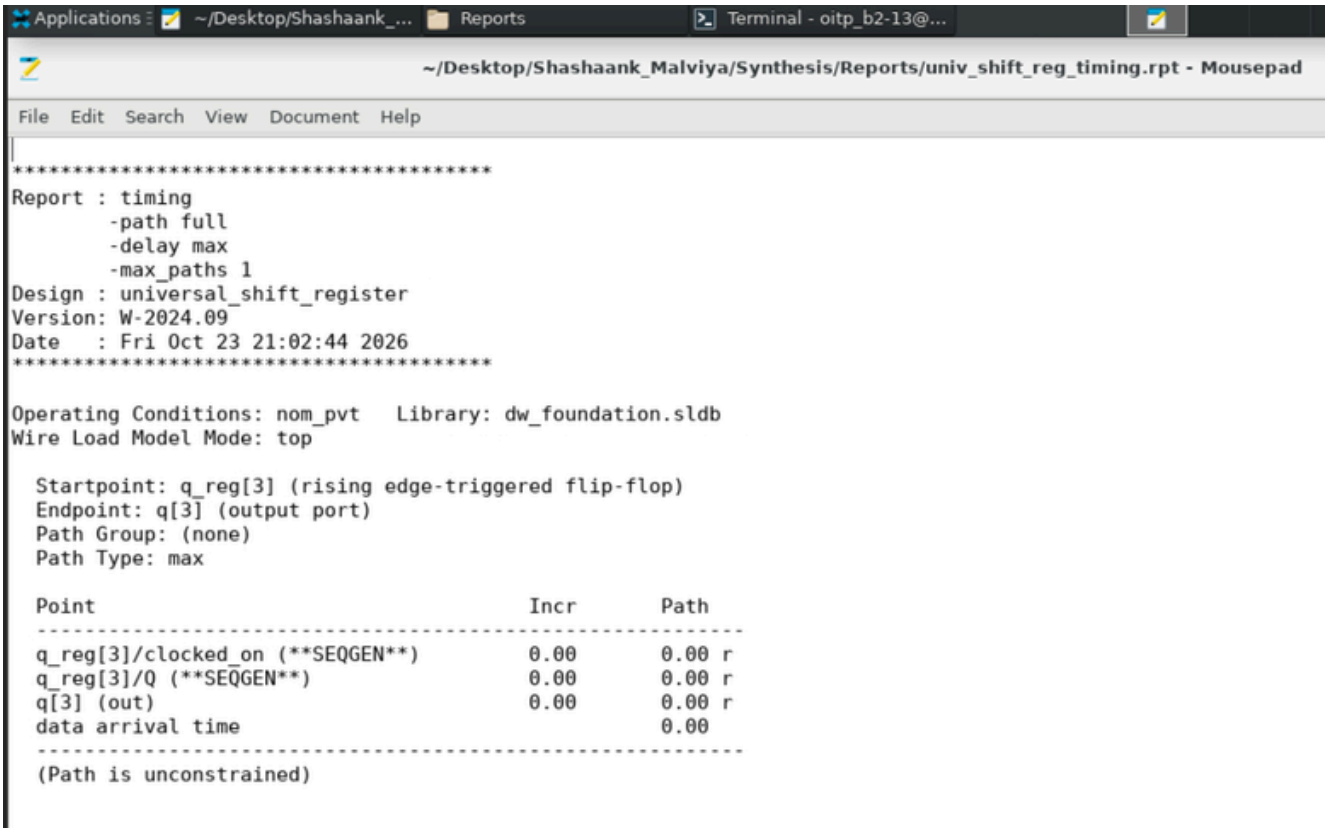

  Cell Internal Power  =   0.0000
  Net Switching Power  =   0.0000
                         ---------
Total Dynamic Power    =   0.0000

Cell Leakage Power     =   0.0000

Error: Either dynamic power or leakage power, in the library, is unitless. Unable to display complete power group summary. (PWR-799)
Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000                NA   (     N/A)
memory             0.0000           0.0000            0.0000                NA   (     N/A)
black_box          0.0000           0.0000            0.0000                NA   (     N/A)
clock_network      0.0000           0.0000            0.0000                NA   (     N/A)  i
register           0.0000           0.0000            0.0000                NA   (     N/A)
sequential         0.0000           0.0000            0.0000                NA   (     N/A)
combinational      0.0000           0.0000            0.0000                NA   (     N/A)
--------------------------------------------------------------------------------------------------
Total              0.0000           0.0000            0.0000                NA
1
```

- ## Timing Report:



Applications ~/Desktop/Shashaank_... Reports Terminal - oitp_b2-13@...

~/Desktop/Shashaank_Malviya/Synthesis/Reports/univ_shift_reg_timing.rpt - Mousepad

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```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : universal_shift_register
Version: W-2024.09
Date   : Fri Oct 23 21:02:44 2026
****************************************

Operating Conditions: nom_pvt   Library: dw_foundation.sldb
Wire Load Model Mode: top

  Startpoint: q_reg[3] (rising edge-triggered flip-flop)
  Endpoint: q[3] (output port)
  Path Group: (none)
  Path Type: max

  Point                                    Incr       Path
  -----------------------------------------------------------
  q_reg[3]/clocked_on (**SEQGEN**)         0.00       0.00 r
  q_reg[3]/Q (**SEQGEN**)                  0.00       0.00 r
  q[3] (out)                               0.00       0.00 r
  data arrival time                                   0.00
  -----------------------------------------------------------
  (Path is unconstrained)
```

- ## Area Report:

```
Area
-----------------------------------
Combinational Area:        0.000000
Noncombinational Area:     0.000000
Buf/Inv Area:              0.000000
Total Buffer Area:             0.00
Total Inverter Area:           0.00
Macro/Black Box Area:      0.000000
Net Area:                  0.000000
-----------------------------------
Cell Area:                 0.000000
Design Area:               0.000000


Design Rules
-----------------------------------
Total Number of Nets:             27
Nets With Violations:              0
Max Trans Violations:              0
Max Cap Violations:                0
-----------------------------------


Hostname: smartlab6.calicut.nielit.in

Compile CPU Statistics
-----------------------------------------
Resource Sharing:                   0.00
Logic Optimization:                 0.00
```

Q2. Design and synthesize an 8-bit Arithmetic Logic Unit (ALU) supporting arithmetic and logical operations.

a) Implement the complete functional specification of the ALU.

b) Analyze the synthesized implementation using the generated design reports.

c) Examine the technology mapped netlist and identify the major standard cells utilized.

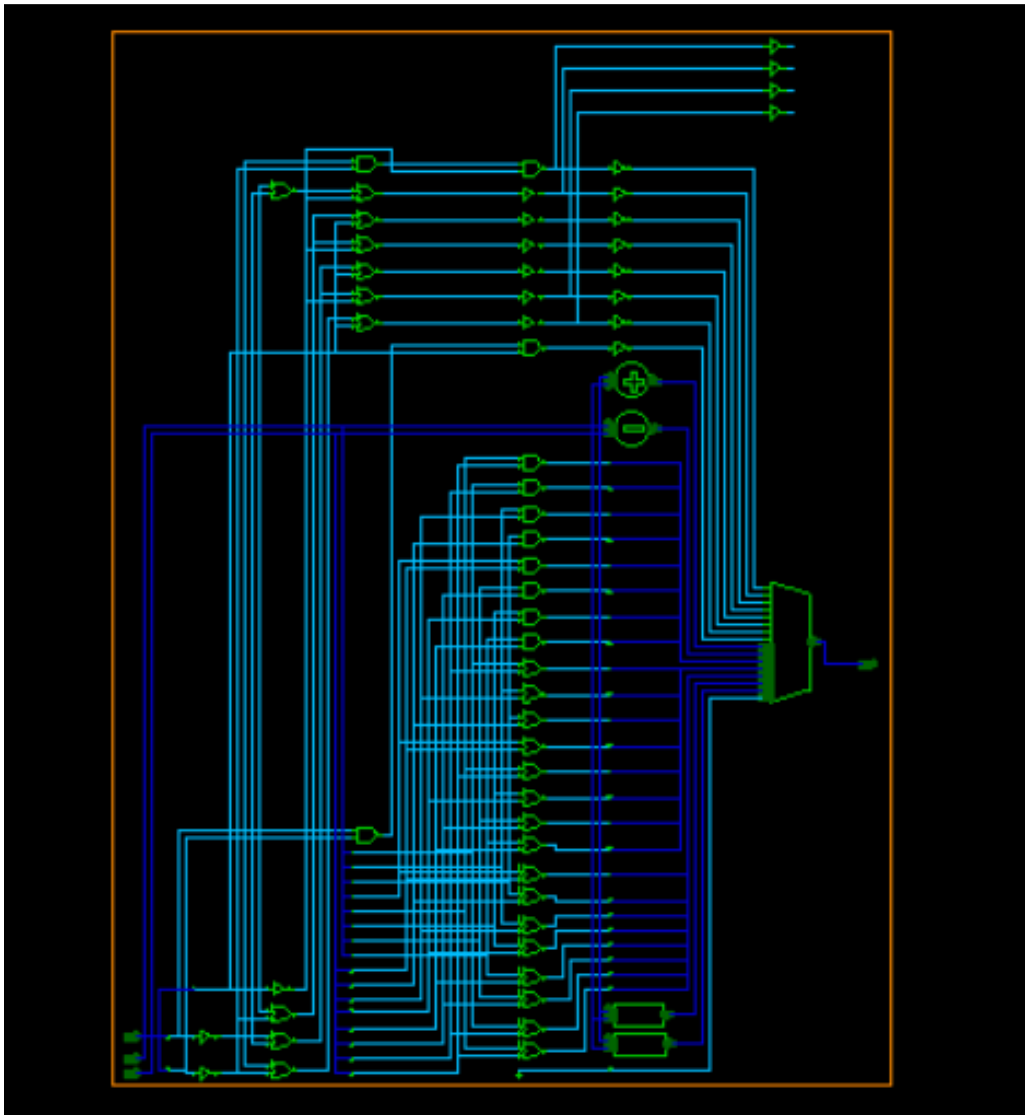
d) Evaluate the quality of the synthesized design based on the obtained implementation results.

- ## Testbench:

```
`timescale 1ns/1ps
module tb_alu_8bit;
    reg [7:0] A, B;
    reg [2:0] opcode;
    wire [7:0] result;
    alu_8bit uut (
        .A(A),
        .B(B),
        .opcode(opcode),
        .result(result)
    );
    initial begin
        $dumpfile("../VCD/alu_8bit.vcd");
        $dumpvars(0, tb_alu_8bit);
        // Addition:
        A = 8'd20; B = 8'd10; opcode = 3'b000;
        #10;
        // Subtraction:
        A = 8'd20; B = 8'd10; opcode = 3'b001;
        #10;
        // AND:
        A = 8'hF0; B = 8'h0F; opcode = 3'b010;
        #10;
        // OR:
        A = 8'hF0; B = 8'h0F; opcode = 3'b011;
        #10;
        // XOR:
        A = 8'hAA; B = 8'hFF; opcode = 3'b100;
        #10;
        // Left shift:
        A = 8'h0F; B = 8'd2; opcode = 3'b101;
        #10;
        // Right shift:
        A = 8'hF0; B = 8'd2; opcode = 3'b110;
        #10;
        $finish;
```

```
end
initial begin
$monitor("Time=%0t A=%h B=%h opcode=%b result=%h",
$time, A, B, opcode, result);
end
endmodule
```

- ## Schematic:



- ## Commands:

vcs -o Output/alu_8bit.out ../RTL/Design/alu_8bit.v ../RTL/TB/tb_alu_8bit.v

dc_shell

source synth.tcl

gui_show

- **sdc file constraint: create_clock -name virtual_clk -period 10**

set_input_delay 1 -clock virtual_clk [all_inputs]
set_output_delay 1 -clock virtual_clk [all_outputs]

- **Netlist:**

```
////////////////////////////////////////////////////////////
// Created by: Synopsys Design Compiler(R)
// Version   : W-2024.09
// Date      : Fri Oct 23 21:17:45 2026
////////////////////////////////////////////////////////////


module alu_8bit ( A, B, opcode, result );
  input [7:0] A;
  input [7:0] B;
  input [2:0] opcode;
  output [7:0] result;
  wire   N0, N1, N2, N3, N4, N5, N6, N7, N8, N9, N10, N11, N12, N13, N14, N15,
        N16, N17, N18, N19, N20, N21, N22, N23, N24, N25, N26, N27, N28, N29,
        N30, N31, N32, N33, N34, N35, N36, N37, N38, N39, N40, N41, N42, N43,
        N44, N45, N46, N47, N48, N49, N50, N51, N52, N53, N54, N55, N56, N57,
        N58, N59, N60, N61, N62, N63, N64, N65, N66, N67, N68, N69, N70, N71,
        N72, N73, N74, N75, N76, N77, N78, N79, N80, N81, N82, N83, N84, N85,
        N86;

  GTECH_AND2 C7 ( .A(N20), .B(N13), .Z(N8) );
  GTECH_AND2 C8 ( .A(N8), .B(N17), .Z(N9) );
  GTECH_OR2 C10 ( .A(opcode[2]), .B(opcode[1]), .Z(N10) );
  GTECH_OR2 C11 ( .A(N10), .B(N17), .Z(N11) );
  GTECH_OR2 C14 ( .A(opcode[2]), .B(N13), .Z(N14) );
  GTECH_OR2 C15 ( .A(N14), .B(opcode[0]), .Z(N15) );
  GTECH_OR2 C20 ( .A(N14), .B(N17), .Z(N18) );
  GTECH_OR2 C23 ( .A(N20), .B(opcode[1]), .Z(N21) );
  GTECH_OR2 C24 ( .A(N21), .B(opcode[0]), .Z(N22) );
  GTECH_OR2 C29 ( .A(N21), .B(N17), .Z(N24) );
  GTECH_OR2 C33 ( .A(N20), .B(N13), .Z(N26) );
  GTECH_OR2 C34 ( .A(N26), .B(opcode[0]), .Z(N27) );
  GTECH_AND2 C36 ( .A(opcode[2]), .B(opcode[1]), .Z(N29) );
  GTECH_AND2 C37 ( .A(N29), .B(opcode[0]), .Z(N30) );
  ASH_UNS_UNS_OP sll_16 ( .A(A), .SH(B[2:0]), .Z({N78, N77, N76, N75, N74, N73,
      N72, N71}) );
  ASHR_UNS_UNS_OP srl_17 ( .A(A), .SH(B[2:0]), .Z({N86, N85, N84, N83, N82,
      N81, N80, N79}) );
  ADD_UNS_OP add_11 ( .A(A), .B(B), .Z({N38, N37, N36, N35, N34, N33, N32, N31}) );
  SUB_UNS_OP sub_12 ( .A(A), .B(B), .Z({N46, N45, N44, N43, N42, N41, N40, N39}) );
  SELECT_OP C87 ( .DATA1({N38, N37, N36, N35, N34, N33, N32, N31}), .DATA2({
      N46, N45, N44, N43, N42, N41, N40, N39}), .DATA3({N47, N48, N49, N50,
      N51, N52, N53, N54}), .DATA4({N55, N56, N57, N58, N59, N60, N61, N62}),
      .DATA5({N63, N64, N65, N66, N67, N68, N69, N70}), .DATA6({N78, N77,
      N76, N75, N74, N73, N72, N71}), .DATA7({N86, N85, N84, N83, N82, N81,
      N80, N79}), .DATA8({1'b0, 1'b0, 1'b0, 1'b0, 1'b0, 1'b0, 1'b0, 1'b0}),
      .CONTROL1(N0), .CONTROL2(N1), .CONTROL3(N2), .CONTROL4(N3), .CONTROL5(
      N4), .CONTROL6(N5), .CONTROL7(N6), .CONTROL8(N7), .Z(result) );
```

```
 GTECH_BUF B_0 ( .A(N9), .Z(N0) );
  GTECH_BUF B_1 ( .A(N12), .Z(N1) );
  GTECH_BUF B_2 ( .A(N16), .Z(N2) );
  GTECH_BUF B_3 ( .A(N19), .Z(N3) );
  GTECH_BUF B_4 ( .A(N23), .Z(N4) );
  GTECH_BUF B_5 ( .A(N25), .Z(N5) );
  GTECH_BUF B_6 ( .A(N28), .Z(N6) );
  GTECH_BUF B_7 ( .A(N30), .Z(N7) );
  GTECH_NOT I_0 ( .A(N11), .Z(N12) );
  GTECH_NOT I_1 ( .A(opcode[1]), .Z(N13) );
  GTECH_NOT I_2 ( .A(N15), .Z(N16) );
  GTECH_NOT I_3 ( .A(opcode[0]), .Z(N17) );
  GTECH_NOT I_4 ( .A(N18), .Z(N19) );
  GTECH_NOT I_5 ( .A(opcode[2]), .Z(N20) );
  GTECH_NOT I_6 ( .A(N22), .Z(N23) );
  GTECH_NOT I_7 ( .A(N24), .Z(N25) );
  GTECH_NOT I_8 ( .A(N27), .Z(N28) );
  GTECH_BUF B_8 ( .A(N9) );
  GTECH_BUF B_9 ( .A(N12) );
  GTECH_AND2 C109 ( .A(A[7]), .B(B[7]), .Z(N47) );
  GTECH_AND2 C110 ( .A(A[6]), .B(B[6]), .Z(N48) );
  GTECH_AND2 C111 ( .A(A[5]), .B(B[5]), .Z(N49) );
  GTECH_AND2 C112 ( .A(A[4]), .B(B[4]), .Z(N50) );
  GTECH_AND2 C113 ( .A(A[3]), .B(B[3]), .Z(N51) );
  GTECH_AND2 C114 ( .A(A[2]), .B(B[2]), .Z(N52) );
  GTECH_AND2 C115 ( .A(A[1]), .B(B[1]), .Z(N53) );
  GTECH_AND2 C116 ( .A(A[0]), .B(B[0]), .Z(N54) );
  GTECH_OR2 C117 ( .A(A[7]), .B(B[7]), .Z(N55) );
  GTECH_OR2 C118 ( .A(A[6]), .B(B[6]), .Z(N56) );
  GTECH_OR2 C119 ( .A(A[5]), .B(B[5]), .Z(N57) );
  GTECH_OR2 C120 ( .A(A[4]), .B(B[4]), .Z(N58) );
  GTECH_OR2 C121 ( .A(A[3]), .B(B[3]), .Z(N59) );
  GTECH_OR2 C122 ( .A(A[2]), .B(B[2]), .Z(N60) );
  GTECH_OR2 C123 ( .A(A[1]), .B(B[1]), .Z(N61) );
  GTECH_OR2 C124 ( .A(A[0]), .B(B[0]), .Z(N62) );
  GTECH_XOR2 C125 ( .A(A[7]), .B(B[7]), .Z(N63) );
  GTECH_XOR2 C126 ( .A(A[6]), .B(B[6]), .Z(N64) );
  GTECH_XOR2 C127 ( .A(A[5]), .B(B[5]), .Z(N65) );
  GTECH_XOR2 C128 ( .A(A[4]), .B(B[4]), .Z(N66) );
  GTECH_XOR2 C129 ( .A(A[3]), .B(B[3]), .Z(N67) );
  GTECH_XOR2 C130 ( .A(A[2]), .B(B[2]), .Z(N68) );
  GTECH_XOR2 C131 ( .A(A[1]), .B(B[1]), .Z(N69) );
  GTECH_XOR2 C132 ( .A(A[0]), .B(B[0]), .Z(N70) );
  GTECH_BUF B_10 ( .A(N25) );
  GTECH_BUF B_11 ( .A(N28) );
endmodule
```

- **Power Report:**

```
File  Edit  Search  View  Document  Help
    Time Units = Unitless
    Dynamic Power Units = Unitless    (derived from V,C,T units)
    Leakage Power Units = Unitless


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =   0.0000
  Net Switching Power  =   0.0000
                         ---------
Total Dynamic Power    =   0.0000

Cell Leakage Power     =   0.0000

Error: Either dynamic power or leakage power, in the library, is unitless. Unable to display complete power group summary. (PWR-799)
Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000               NA   (     N/A)
memory             0.0000           0.0000            0.0000               NA   (     N/A)
black_box          0.0000           0.0000            0.0000               NA   (     N/A)
clock_network      0.0000           0.0000            0.0000               NA   (     N/A)  i
register           0.0000           0.0000            0.0000               NA   (     N/A)
sequential         0.0000           0.0000            0.0000               NA   (     N/A)
combinational      0.0000           0.0000            0.0000               NA   (     N/A)
--------------------------------------------------------------------------------------------------
Total              0.0000           0.0000            0.0000               NA
1
```

- **Timing Report:**

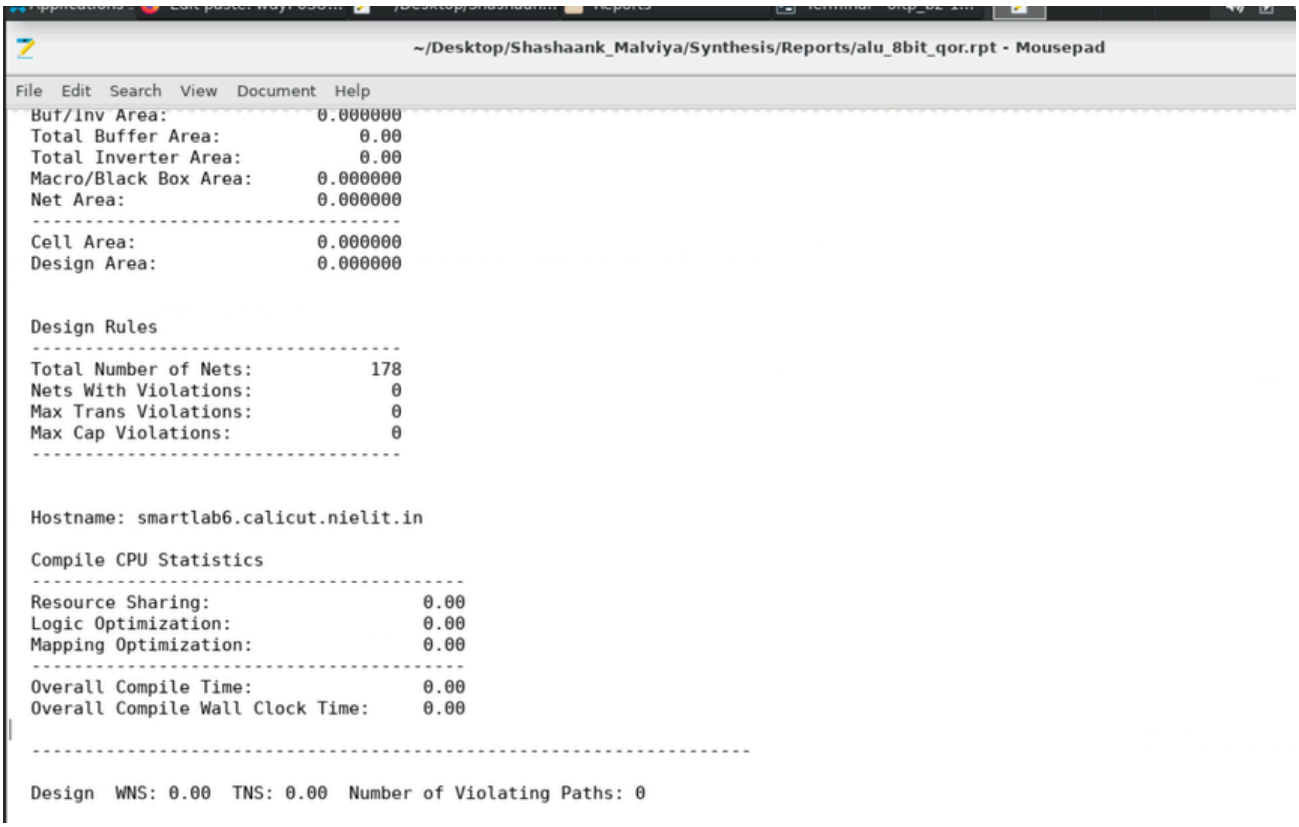
```
~/Desktop/Shashaank_Malviya/Synthesis/Reports/alu_8bit_timing.rpt
File  Edit  Search  View  Document  Help
Wire Load Model Mode: top

  Startpoint: A[0] (input port clocked by virtual_clk)
  Endpoint: result[0] (output port clocked by virtual_clk)
  Path Group: virtual_clk
  Path Type: max

  Point                                                   Incr       Path
  ---------------------------------------------------------------------------
  clock virtual_clk (rise edge)                           0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  input external delay                                    1.00       1.00 r
  A[0] (in)                                               0.00       1.00 r
  add_11/A_0 (*ADD_UNS_OP_8_8_8)                          0.00       1.00 r
  add_11/*cell*4/A[0] (DW01_add_width8)                   0.00       1.00 r
  ...
  add_11/*cell*4/SUM[0] (DW01_add_width8)                 0.00       1.00 r
  add_11/Z_0 (*ADD_UNS_OP_8_8_8)                          0.00       1.00 r
  C87/Z_0 (*SELECT_OP_8.8_8.1_8)                          0.00       1.00 r
  result[0] (out)                                         0.00       1.00 r
  data arrival time                                                  1.00

  clock virtual_clk (rise edge)                          10.00      10.00
  clock network delay (ideal)                             0.00      10.00
  output external delay                                  -1.00       9.00
  data required time                                                 9.00
  ---------------------------------------------------------------------------
  data required time                                                 9.00
  data arrival time                                                 -1.00
  ---------------------------------------------------------------------------
  slack (MET)                                                        8.00
```

- ## Area Report:



Q3. Design and synthesize a Sequence Detector for detecting the binary sequence 10110.
a) Develop an appropriate finite state machine for the required sequence.
b) Perform logic synthesis and validate the synthesized implementation.
c) Examine the technology mapped schematic and synthesized design hierarchy.
d) Analyze the generated synthesis reports and discuss the implementation characteristics of the synthesized design.

- ## Commands:

vcs -o Output/seq_det.out ../RTL/Design/seq_det.v ../RTL/TB/tb_seq_det.v

gtkwave ../VCD/seq_det.vcd

dc_shell

source synth.tcl

gui_show

- **sdc file constraint:**

create_clock -name clk -period 10 [get_ports clk]
set_input_delay 1 -clock clk [get_ports din]
set_output_delay 1 -clock clk [get_ports detected]

- **Module:**

```
`timescale 1ns/1ps
module seq_det (
    input wire clk,
    input wire rst,
    input wire din,
    output reg detected
);
    localparam S0 = 3'b000,
               S1 = 3'b001,
               S2 = 3'b010,
               S3 = 3'b011,
               S4 = 3'b100;

    reg [2:0] state, next_state;

    // State register
    always @(posedge clk or posedge rst) begin
        if (rst)
            state <= S0;
        else
            state <= next_state;
    end

    // Next-state logic
    always @(*) begin
        case (state)
            S0: next_state = din ? S1 : S0;
            S1: next_state = din ? S1 : S2;
            S2: next_state = din ? S3 : S0;
            S3: next_state = din ? S4 : S2;
            S4: next_state = din ? S1 : S2;
            default: next_state = S0;
        endcase
    end

    // Mealy output logic
    always @(*) begin
        detected = (state == S4) && (din == 1'b0);
    end
endmodule
```

- **Testbench:**

```
`timescale 1ns/1ps
module tb_seq_det;

    reg clk, rst, din;
    wire detected;

    seq_det uut (
        .clk(clk),
        .rst(rst),
        .din(din),
        .detected(detected)
    );

    always #5 clk = ~clk;
    task send_bit;
        input bit_value;
        begin
            @(negedge clk);
            din = bit_value;
            #1;
            $display("Time=%0t din=%b detected=%b",
                    $time, din, detected);
        end
    endtask

    initial begin
        $dumpfile("../VCD/seq_det.vcd");
        $dumpvars(0, tb_seq_det);

        clk = 0;
        rst = 1;
        din = 0;
        #12;
        rst = 0;

        // Input 10110110:
        send_bit(1);
        send_bit(0);
        send_bit(1);
        send_bit(1);
        send_bit(0);

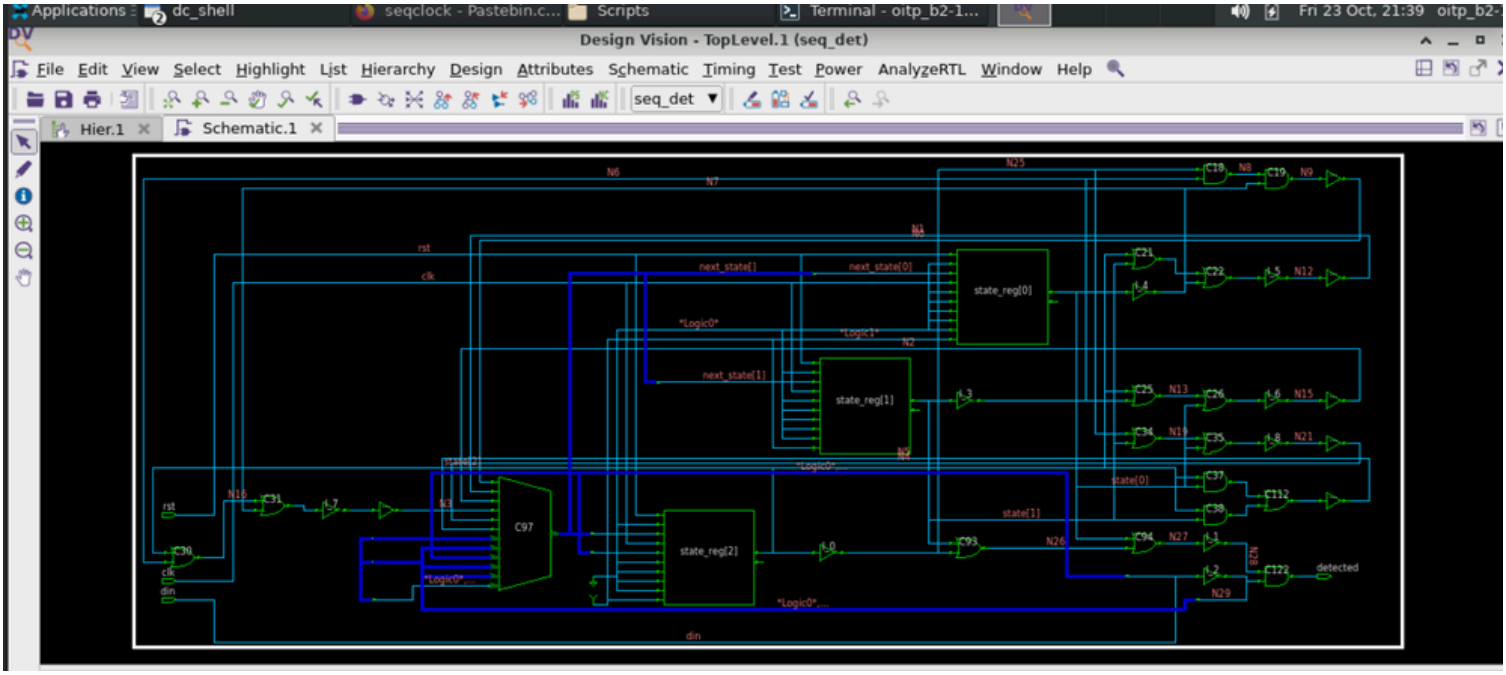
        send_bit(1);
        send_bit(1);
        send_bit(0);
```

```
        // Additional test bits
        send_bit(0);
        send_bit(1);
        send_bit(0);

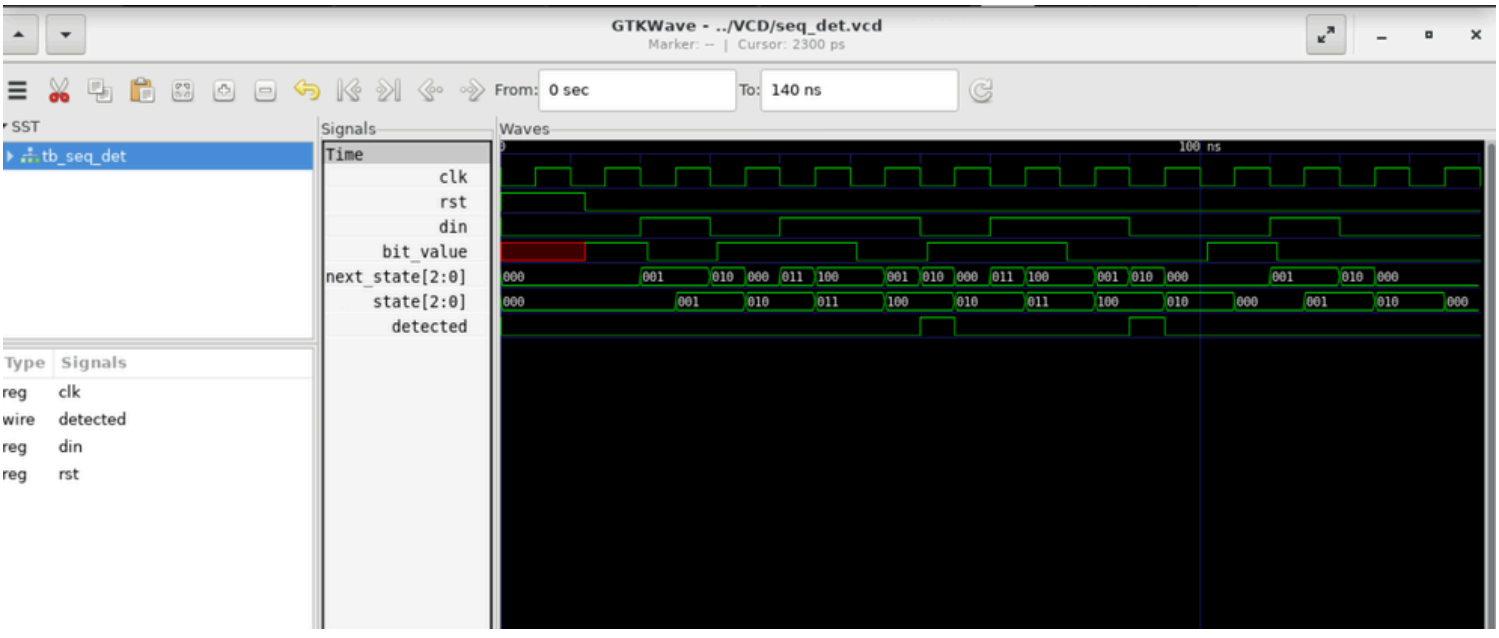
        repeat (2) @(negedge clk);
        $finish;
    end

endmodule
```

- **Schematic:**



- **Output Waveform:**



- **Netlist:**

```
////////////////////////////////////////////////////////////
// Created by: Synopsys Design Compiler(R)
// Version   : W-2024.09
// Date      : Fri Oct 23 21:39:01 2026
////////////////////////////////////////////////////////////

module seq_det ( clk, rst, din, detected );
  input clk, rst, din;
  output detected;
  wire   N0, N1, N2, N3, N4, N5, N6, N7, N8, N9, N10, N11, N12, N13, N14, N15,
       N16, N17, N18, N19, N20, N21, N22, N23, N24, N25, N26, N27, N28, N29;
  wire   [2:0] state;
  wire   [2:0] next_state;

  \**SEQGEN**  \state_reg[2]  ( .clear(rst), .preset(1'b0), .next_state(
       next_state[2]), .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(
       state[2]), .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0), .synch_enable(1'b1) );
  \**SEQGEN**  \state_reg[1]  ( .clear(rst), .preset(1'b0), .next_state(
       next_state[1]), .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(
       state[1]), .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0), .synch_enable(1'b1) );
  \**SEQGEN**  \state_reg[0]  ( .clear(rst), .preset(1'b0), .next_state(
       next_state[0]), .clocked_on(clk), .data_in(1'b0), .enable(1'b0), .Q(
       state[0]), .synch_clear(1'b0), .synch_preset(1'b0), .synch_toggle(1'b0), .synch_enable(1'b1) );
  GTECH_AND2 C18 ( .A(N25), .B(N6), .Z(N8) );
  GTECH_AND2 C19 ( .A(N8), .B(N7), .Z(N9) );
  GTECH_OR2 C21 ( .A(state[2]), .B(state[1]), .Z(N10) );
  GTECH_OR2 C22 ( .A(N10), .B(N7), .Z(N11) );
  GTECH_OR2 C25 ( .A(state[2]), .B(N6), .Z(N13) );
  GTECH_OR2 C26 ( .A(N13), .B(state[0]), .Z(N14) );
  GTECH_OR2 C30 ( .A(state[2]), .B(N6), .Z(N16) );
  GTECH_OR2 C31 ( .A(N16), .B(N7), .Z(N17) );
  GTECH_OR2 C34 ( .A(N25), .B(state[1]), .Z(N19) );
  GTECH_OR2 C35 ( .A(N19), .B(state[0]), .Z(N20) );
  GTECH_AND2 C37 ( .A(state[2]), .B(state[0]), .Z(N22) );
  GTECH_AND2 C38 ( .A(state[2]), .B(state[1]), .Z(N23) );
  GTECH_NOT I_0 ( .A(state[2]), .Z(N25) );
  GTECH_OR2 C93 ( .A(state[1]), .B(N25), .Z(N26) );
  GTECH_OR2 C94 ( .A(state[0]), .B(N26), .Z(N27) );
  GTECH_NOT I_1 ( .A(N27), .Z(N28) );
  GTECH_NOT I_2 ( .A(din), .Z(N29) );
  SELECT_OP C97 ( .DATA1({1'b0, 1'b0, din}), .DATA2({1'b0, N29, din}), .DATA3(
       {1'b0, din, din}), .DATA4({din, N29, 1'b0}), .DATA5({1'b0, N29, din}),
       .DATA6({1'b0, 1'b0, 1'b0}), .CONTROL1(N0), .CONTROL2(N1), .CONTROL3(N2),
.CONTROL4(N3), .CONTROL5(N4), .CONTROL6(N5), .Z(next_state) );
  GTECH_BUF B_0 ( .A(N9), .Z(N0) );
  GTECH_BUF B_1 ( .A(N12), .Z(N1) );
  GTECH_BUF B_2 ( .A(N15), .Z(N2) );
  GTECH_BUF B_3 ( .A(N18), .Z(N3) );
  GTECH_BUF B_4 ( .A(N21), .Z(N4) );
  GTECH_BUF B_5 ( .A(N24), .Z(N5) );
  GTECH_NOT I_3 ( .A(state[1]), .Z(N6) );
  GTECH_NOT I_4 ( .A(state[0]), .Z(N7) );
  GTECH_NOT I_5 ( .A(N11), .Z(N12) );
  GTECH_NOT I_6 ( .A(N14), .Z(N15) );
  GTECH_NOT I_7 ( .A(N17), .Z(N18) );
  GTECH_NOT I_8 ( .A(N20), .Z(N21) );
  GTECH_OR2 C112 ( .A(N22), .B(N23), .Z(N24) );
  GTECH_AND2 C122 ( .A(N28), .B(N29), .Z(detected) );
endmodule
```

- **Power Report:**

~/Desktop/Shashaank_Malviya/Synthesis/Reports/seq_det_power.rpt - Mousepad

File Edit Search View Document Help

univ_shift_reg_synth_netlist.v | seq_det_power.rpt

```
    Dynamic Power Units = Unitless    (derived from V,C,T units)
    Leakage Power Units = Unitless


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =   0.0000
  Net Switching Power  =   0.0000
                         ---------
Total Dynamic Power    =   0.0000

Cell Leakage Power     =   0.0000

Error: Either dynamic power or leakage power, in the library, is unitless. Unable to display complete power group summary

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    ) Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000               NA   (     N/A)
memory             0.0000           0.0000            0.0000               NA   (     N/A)
black_box          0.0000           0.0000            0.0000               NA   (     N/A)
clock_network      0.0000           0.0000            0.0000               NA   (     N/A)  i
register           0.0000           0.0000            0.0000               NA   (     N/A)
sequential         0.0000           0.0000            0.0000               NA   (     N/A)
combinational      0.0000           0.0000            0.0000               NA   (     N/A)
--------------------------------------------------------------------------------------------------
Total              0.0000           0.0000            0.0000               NA
1
```

- **Timing Report:**

~/Desktop/Shashaank_Malviya/Synthesis/Reports/seq_det_timi

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```
****************************************

Operating Conditions: nom_pvt   Library: dw_foundation.sldb
Wire Load Model Mode: top

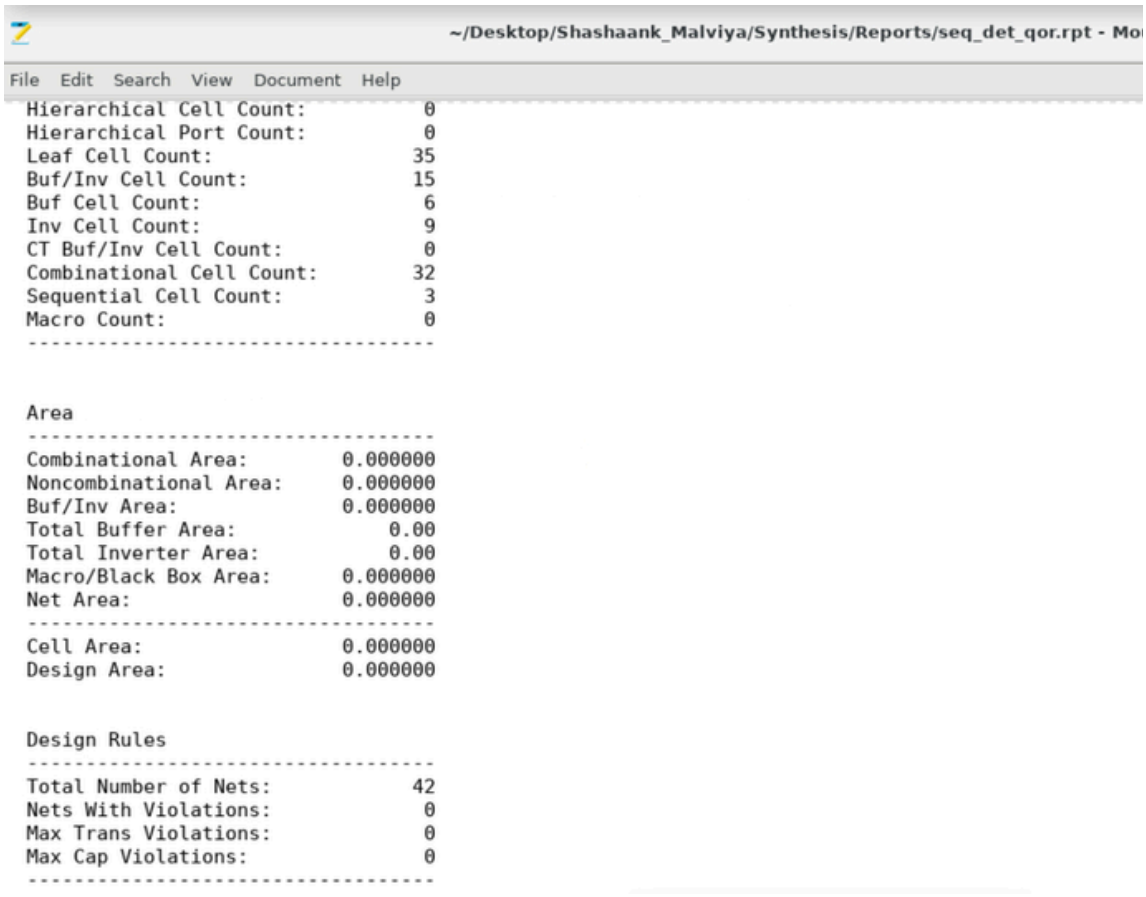
  Startpoint: din (input port clocked by clk)
  Endpoint: detected (output port clocked by clk)
  Path Group: clk
  Path Type: max

  Point                                    Incr       Path
  -----------------------------------------------------------
  clock clk (rise edge)                    0.00       0.00
  clock network delay (ideal)              0.00       0.00
  input external delay                     1.00       1.00 f
  din (in)                                 0.00       1.00 f
  I_2/Z (GTECH_NOT)                        0.00       1.00 r
  C122/Z (GTECH_AND2)                      0.00       1.00 r
  detected (out)                           0.00       1.00 r
  data arrival time                                   1.00

  clock clk (rise edge)                   10.00      10.00
  clock network delay (ideal)              0.00      10.00
  output external delay                   -1.00       9.00
  data required time                                  9.00
  -----------------------------------------------------------
  data required time                                  9.00
  data arrival time                                  -1.00
  -----------------------------------------------------------
  slack (MET)                                         8.00


1
```

- **Area Report:**


~/Desktop/Shashaank_Malviya/Synthesis/Reports/seq_det_qor.rpt - Mo
File Edit Search View Document Help
Hierarchical Cell Count: 0
Hierarchical Port Count: 0
Leaf Cell Count: 35
Buf/Inv Cell Count: 15
Buf Cell Count: 6
Inv Cell Count: 9
CT Buf/Inv Cell Count: 0
Combinational Cell Count: 32
Sequential Cell Count: 3
Macro Count: 0
Area
Combinational Area: 0.000000
Noncombinational Area: 0.000000
Buf/Inv Area: 0.000000
Total Buffer Area: 0.00
Total Inverter Area: 0.00
Macro/Black Box Area: 0.000000
Net Area: 0.000000
Cell Area: 0.000000
Design Area: 0.000000
Design Rules
Total Number of Nets: 42
Nets With Violations: 0
Max Trans Violations: 0
Max Cap Violations: 0